TRIUNE BIBLICAL UNIVERSITY GLOBAL EXTENSION USA

Bachelor of Artificial Intelligence and Robotics Engineering

# AI.309

## Multi-Robot Systems

### COURSE OVERVIEW

Coordinating teams of robots: task allocation, consensus, communication, and collective decision-making.

1

SESSION

60 MINUTES

# ORIENTATION: TEAMS OF ROBOTS

**KEY TOPICS**

- Multi-robot vs. swarm
- Coordination challenges
- Centralized vs. decentralized
- Applications

**TEACHING FOCUS**

Make robots work as a team.

**KEY TAKEAWAY** Teams outperform individuals on complex tasks.

2

SESSION

60 MINUTES

# COMMUNICATION & COORDINATION

**KEY TOPICS**

- Inter-robot communication
- Consensus algorithms
- Leader election
- Formation control

**TEACHING FOCUS**

Align robots on shared goals.

**KEY TAKEAWAY** Coordination requires shared information.

3

SESSION

60 MINUTES

# TASK ALLOCATION

**KEY TOPICS**

- Contract net protocol
- Auction-based allocation
- Optimization methods
- Dynamic repartitioning

**TEACHING FOCUS**

Assign tasks to the right robots.

**KEY TAKEAWAY** Efficient allocation maximizes team performance.

4

SESSION

60 MINUTES

# COOPERATIVE MANIPULATION

**KEY TOPICS**

- Joint carrying
- Coordination of multiple arms
- Force synchronization
- Case studies

**TEACHING FOCUS**

Move large objects together.

**KEY TAKEAWAY** Cooperation handles heavy or awkward loads.

5

SESSION

60 MINUTES

# DECENTRALIZED DECISION MAKING

**KEY TOPICS**

- Distributed optimization
- Game theory for robots
- Voting and consensus
- Scalable protocols

**TEACHING FOCUS**

Decide without a central commander.

**▲ KEY TAKEAWAY** Decentralized teams are resilient.

6

SESSION

60 MINUTES

# MULTI-ROBOT SYSTEMS & REVIEW

**KEY TOPICS**

- Team demo
- Course recap and assessment
- Toward localization (AI.310)
- Team performance eval

**TEACHING FOCUS**

Evaluate a robot team end-to-end.

▲ **KEY TAKEAWAY** A team is judged by its collective result.